Six periodic occupation configurations

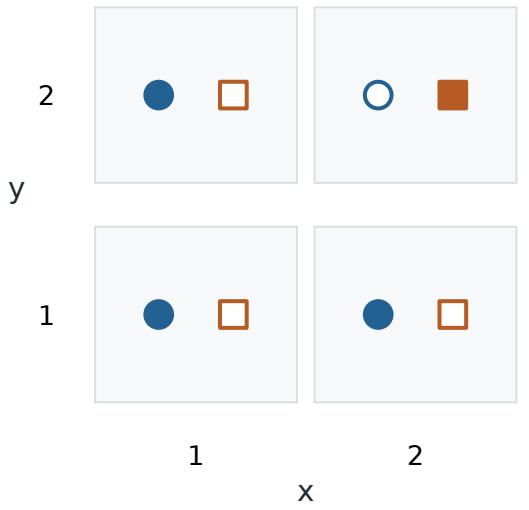
Half filling: one fermion per cell on average

A occupied A empty B occupied B empty

Class 1 | One fermion in every cell

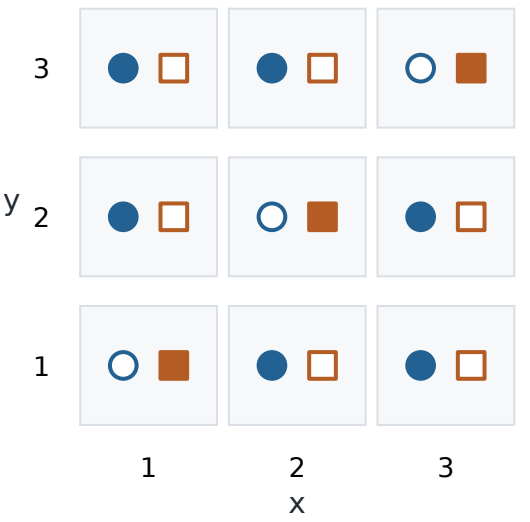
one_B_2

m = 2 | (n0, n1, n2) = (0, 4, 0)



diagonal_B_3

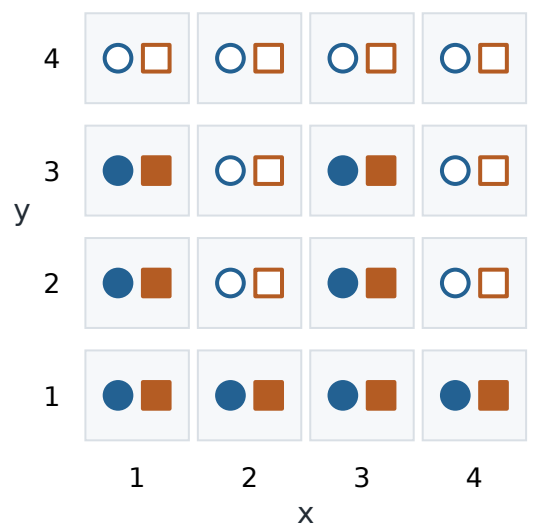
m = 3 | (n0, n1, n2) = (0, 9, 0)



Class 2 | Only empty and doubly occupied cells

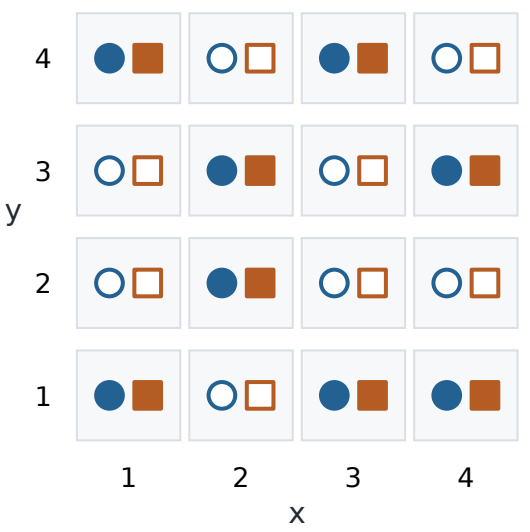
pure02_block4

m = 4 | (n0, n1, n2) = (8, 0, 8)



pure02_irregular4

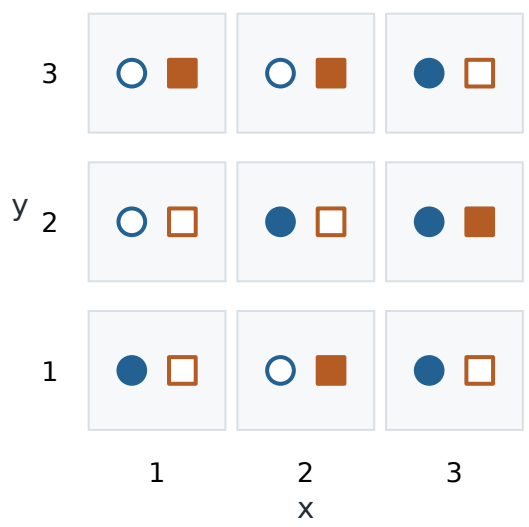
m = 4 | (n0, n1, n2) = (8, 0, 8)



Class 3 | Empty, single and double cells coexist

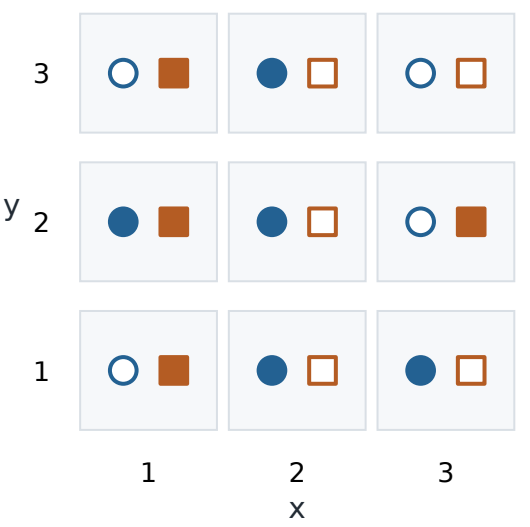
random_half_3

m = 3 | (n0, n1, n2) = (1, 7, 1)



general_matched_a

m = 3 | (n0, n1, n2) = (1, 7, 1)



n0 / n1 / n2: numbers of empty / single / double cells in one periodic tile.

Cell-index diagram: x right, y up; grey box = one A/B cell. Not the real-space honeycomb geometry.